

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 18 — Carboxylic Acids and Derivatives

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	18.1 Carboxylic Acids · Acidity & Inductive Effects · Reactions (Metals, Bases, Carbonates, LiAlH_4) · 18.2 Esters
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	262 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

18.1 Physical Properties & Carboxylate Resonance

— Hydrogen-bonded cyclic dimers in pure state; weak acid dissociation and resonance stabilisation

18.1 Substituent Effects & Chemical Reactions— Negative inductive ($-I$) effect of halogen substituents on acid strength (pK_a); effervescence with NaHCO_3 **18.2 Esters & Hydrolysis Pathways**

— Acid-catalysed condensation esterification (Le Chatelier's equilibrium) vs acyl chloride routes; reversible

SUBTOPIC 18.1 — CARBOXYLIC ACIDS (STRUCTURE, ACIDITY & REACTIONS)

1 Dimer Formation and Carboxylate Resonance — 9701/22/M/J/23/Q3(a)–(d)

HARD · Syllabus 18.1

Carboxylic acids exhibit distinctive physical and chemical properties due to the interaction of the carbonyl and hydroxyl groups. Fig. 1.1 displays the planar hydrogen-bonded cyclic dimer and the resonance stabilisation of the carboxylate anion.

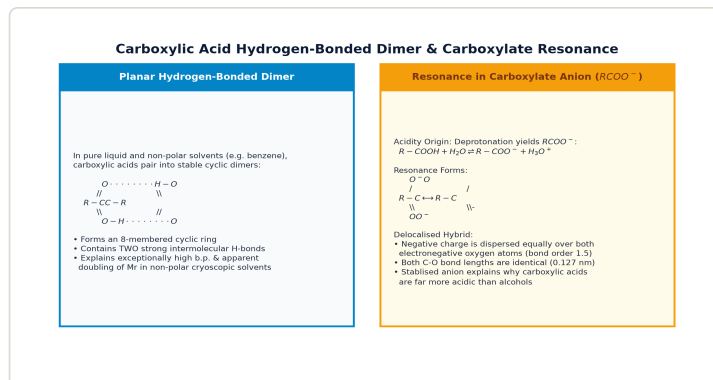


Fig. 1.1: Carboxylic acid cyclic dimer structure and carboxylate resonance delocalisation.

- ((a)) In non-polar solvents such as benzene, the apparent relative molecular mass of ethanoic acid is found to be approximately 120 rather than 60. Explain this observation with reference to intermolecular bonding.

[3]

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- ((b)) Draw the displayed structure of the ethanoate anion (CH_3COO^-) showing how the negative charge is delocalised, and explain why both carbon-oxygen bonds are of equal length.

[3]

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- ((c)) Explain why carboxylic acids are significantly stronger Bronsted-Lowry acids than alcohols, referring to the stability of their respective conjugate bases.

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- ((d)) Write the equation for the acid dissociation of propanoic acid in aqueous solution and give the expression for its acid dissociation constant, K_a .

[2]

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2 Boiling Point Trends and Water Solubility — 9701/21/O/N/23/Q2(a)–(c)

EASY · Syllabus 18.1

Table 2.1 compares the boiling points and water solubilities of four compounds of similar molecular mass ($M_r \approx 72-74$):

- Pentane (b.p. 36°C , insoluble)
- Butanal (b.p. 75°C , moderately soluble)
- Butan-1-ol (b.p. 117°C , soluble)
- Propanoic acid (b.p. 141°C , miscible)

((a)) Explain why propanoic acid has a higher boiling point than both butanal and butan-1-ol. [3]

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((b)) Draw a labelled diagram showing hydrogen bonding between a propanoic acid molecule and a water molecule. [2]

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((c)) Explain why methanoic, ethanoic, and propanoic acids are completely miscible with water, whereas octanoic acid is virtually insoluble. [2]

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3 Synthesis of Carboxylic Acids by Nitrile Hydrolysis — 9701/22/F/M/22/Q3(a)–(c)

HARD · Syllabus 18.1

Carboxylic acids can be synthesised from halogenoalkanes in a two-step sequence involving a nitrile intermediate.

((a)) 1-bromopropane is heated under reflux with ethanolic potassium cyanide. State the IUPAC name and draw the structure of the nitrile formed in Step 1. [2]

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((b)) In Step 2, the nitrile is heated under reflux with dilute hydrochloric acid. Write the balanced equation for this acid hydrolysis. [2]

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((c)) Describe how the hydrolysis could alternatively be carried out in alkaline conditions, and explain how the carboxylic acid would be liberated from the resulting mixture. [3]

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4 Diagnostic Tests: Reactions with Carbonates — 9701/21/M/J/22/Q4(a)–(c)

EASY · Syllabus 18.1

The reaction with sodium carbonate or sodium hydrogencarbonate provides a definitive qualitative diagnostic test for the carboxylic acid group.

((a)) Write the balanced chemical equation for the reaction of ethanoic acid with solid sodium carbonate, Na₂CO₃. [1]

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((b)) State two visible observations during this reaction. [2]

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- ((c)) Describe a confirmatory test on the gas evolved, including the expected observation and equation. [2]
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5 Reaction with Reactive Metals — 9701/23/M/J/23/Q2(a)–(c)

EASY · Syllabus 18.1

Carboxylic acids react with reactive s-block metals.

- ((a)) Write a balanced chemical equation for the reaction of propanoic acid with magnesium ribbon. [1]
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- ((b)) State the visible observations and describe the test to confirm the identity of the gas produced. [2]
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- ((c)) Give the systematic name and structural formula of the organic salt formed. [2]
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6 Neutralisation Titrations and K_a Calculation — 9701/22/M/J/22/Q2(a)–(d)

HARD · Syllabus 18.1

A 25.0 cm³ sample of an unknown monoprotic carboxylic acid, RCOOH, requires 21.40 cm³ of 0.100 mol dm⁻³ NaOH for complete neutralisation. The initial pH of the 25.0 cm³ acid solution was 2.88.

- ((a)) Calculate the concentration of the carboxylic acid in mol dm⁻³. [2]
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- ((b)) Calculate the concentration of hydrogen ions, [H⁺], in the original acid solution. [1]
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- ((c)) Calculate the value of the acid dissociation constant, K_a , for this carboxylic acid, stating its units. [2]
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- ((d)) Select a suitable indicator for this weak acid–strong base titration from phenolphthalein (pH range 8.3–10.0) and methyl orange (pH range 3.1–4.4). Justify your choice. [2]
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7 Reduction of Carboxylic Acids using LiAlH_4 — 9701/21/O/N/22/Q3(a)–(c)

HARD · Syllabus 18.1

Carboxylic acids are resistant to mild reducing agents, but can be reduced by lithium aluminium hydride, LiAlH_4 .

- ((a))** State the reagent, solvent, and write a balanced equation for the reduction of butanoic acid using $[\text{H}]$ for the reducing agent. **[3]**

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- ((b))** Explain why sodium borohydride, NaBH_4 , cannot be used to reduce carboxylic acids, whereas LiAlH_4 is effective. **[2]**

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- ((c))** State why water must be completely excluded from the reaction vessel when using LiAlH_4 . **[1]**

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8 Chlorination to Acyl Chlorides with PCl_5 and SOCl_2 — 9701/22/O/N/23/Q5

HARD · Syllabus 18.1

Carboxylic acids react with halogenating reagents to convert the $-\text{COOH}$ group into an acyl chloride group ($-\text{COCl}$).

- ((a))** Write the balanced equation for the reaction of ethanoic acid with phosphorus(V) chloride, PCl_5 , at room temperature. **[1]**

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- ((b))** Write the balanced equation for the reaction of ethanoic acid with thionyl chloride, SOCl_2 . **[1]**

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- ((c))** Explain why SOCl_2 is the preferred industrial and laboratory reagent for this preparation rather than PCl_5 . **[2]**

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- ((d))** State one visible observation that occurs in both reactions. **[1]**

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9 Dicarboxylic Acids: Ethanedioic Acid Oxidation — 9701/23/O/N/21/Q2(a)–(c)

HARD · Syllabus 18.1

Ethanedioic acid (oxalic acid), $\text{HOOC}-\text{COOH}$, is a dicarboxylic acid that undergoes redox reactions with potassium manganate(VII).

- ((a))** Write the oxidation half-equation for the oxidation of ethanedioic acid to carbon dioxide. **[1]**

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- ((b))** Write the reduction half-equation for acidified manganate(VII) ions, MnO_4^- , in acidic solution. **[1]**

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((c)) Combine the half-equations to produce the full ionic equation for the titration of ethanedioic acid with acidified potassium manganate(VII). **[2]**

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((d)) Explain why standard monocarboxylic acids such as ethanoic acid do NOT react with acidified potassium manganate(VII). **[2]**

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10 Multiple Choice: Formation of Carboxylic Acids — 9701/12/M/J/23/Q26

EASY · Syllabus 18.1

Which reaction does NOT produce a carboxylic acid?

((a)) Select the correct reaction: **[1]**

A Acid hydrolysis of propanenitrile under reflux

B Oxidation of propan-2-ol with acidified $K_2Cr_2O_7$ under reflux

C Oxidation of propan-1-ol with acidified $K_2Cr_2O_7$ under reflux

D Acid hydrolysis of ethyl propanoate under reflux

A

B

C

D

((b)) State the organic product formed in the reaction you selected. **[1]**

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11 Methanoic Acid as a Unique Reducing Carboxylic Acid — 9701/21/M/J/21/Q4

HARD · Syllabus 18.1

Methanoic acid, $HCOOH$, is unique among carboxylic acids because it exhibits reducing properties typical of aldehydes.

((a)) Examine the structure of $HCOOH$ and explain why it displays reactions characteristic of both an aldehyde and a carboxylic acid. **[2]**

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((b)) State the observation when methanoic acid is warmed with Tollens' reagent and write the equation for this reaction. **[2]**

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((c)) State the observation when methanoic acid is warmed with acidified potassium manganate(VII), and write the balanced equation. **[2]**

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12 Deducing Acid Structure from Combustion Analysis — 9701/22/F/M/21/Q2

HARD · Syllabus 18.1

An organic carboxylic acid Z contains 40.0% carbon, 6.7% hydrogen, and 53.3% oxygen by mass. When 0.150 g of acid Z is dissolved in water and titrated, it neutralises 25.0 cm³ of 0.100 mol dm⁻³ NaOH.

((a)) Determine the empirical formula of acid Z. **[2]**

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((b)) Calculate the number of moles of NaOH used and determine the relative molecular mass (*M_r*) of acid Z, assuming Z is a monoprotic acid. **[2]**

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((c)) Deduce the molecular formula and systematic IUPAC name of acid Z. **[2]**

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13 Multiple Choice: Relative Acidities — 9701/11/M/J/23/Q24

EASY · Syllabus 18.1

Which sequence correctly places the compounds in order of decreasing acid strength (strongest first)?

((a)) **[1]**

Select the correct order:

- A Ethanoic acid > Phenol > Ethanol > Water
- B Ethanoic acid > Phenol > Water > Ethanol
- C Phenol > Ethanoic acid > Water > Ethanol
- D Ethanol > Water > Phenol > Ethanoic acid

- A
- B
- C
- D

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((b)) Explain why phenol is more acidic than ethanol but less acidic than ethanoic acid. **[2]**

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14 Effect of Chlorine Substituents on Acid Strength — 9701/22/M/J/23/Q6(a)-(d)

HARD · Syllabus 18.1

The strength of carboxylic acids is significantly altered by substituents on the carbon chain. Fig. 14.1 compares the *pK_a* values of ethanoic acid and its chlorinated derivatives.

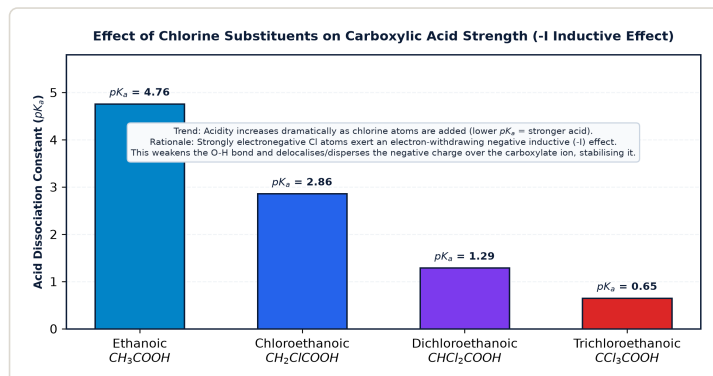


Fig. 14.1: Bar chart comparing pKa values of ethanoic acid through trichloroethanoic acid.

- ((a)) State the trend in acid strength shown in Fig. 14.1 as more chlorine atoms are introduced onto the alpha-carbon. [1]
- ((b)) Explain this trend in terms of the electronegativity of chlorine and the inductive effect on the O-H bond of the undissociated acid. [2]
- ((c)) Explain this trend in terms of the stability of the conjugate base (carboxylate anion). [2]
- ((d)) Predict whether 2-chloropropanoic acid or 3-chloropropanoic acid is the stronger acid. Explain your reasoning in terms of distance from the reaction centre. [3]

15

Comparative Acidity of Halogenated Acids (F vs Cl vs Br vs I) — 9701/21/O/N/23/Q4

HARD · Syllabus 18.1

Table 15.1 lists the pKa values of four monohalogenated ethanoic acids:

- Fluoroethanoic acid (pKa = 2.60)
- Chloroethanoic acid (pKa = 2.86)
- Bromoethanoic acid (pKa = 2.90)
- Iodoethanoic acid (pKa = 3.18)

- ((a)) Explain the trend in pKa values across this series of halogenoethanoic acids. [3]
- ((b)) Calculate the Ka value of fluoroethanoic acid and compare it to that of ethanoic acid (Ka = 1.74 x 10 mol dm³). [2]

- ((c)) Suggest how the pK_a of trifluoroethanoic acid, CF_3COOH , would compare with that of trichloroethanoic acid, CCl_3COOH . [1]
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16 Electron-Donating Alkyl Groups and Acid Strength — 9701/22/O/N/22/Q4

HARD · Syllabus 18.1

Consider the pK_a values of methanoic acid ($pK_a = 3.75$), ethanoic acid ($pK_a = 4.76$), and propanoic acid ($pK_a = 4.87$).

- ((a)) Explain why methanoic acid is significantly stronger than ethanoic acid. [2]
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- ((b)) Explain why propanoic acid is slightly weaker than ethanoic acid. [2]
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- ((c)) Predict whether 2,2-dimethylpropanoic acid, $(CH_3)_3CCOOH$, is a stronger or weaker acid than ethanoic acid. Justify your answer. [2]
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17 Acid-Base Neutralisation Curves — 9701/21/M/J/23/Q5(a)-(c)

HARD · Syllabus 18.1

Fig. 17.1 illustrates the pH titration curve obtained when 20.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ chloroethanoic acid is titrated with $0.100 \text{ mol dm}^{-3}$ NaOH.

- ((a)) Explain why the initial pH of $0.100 \text{ mol dm}^{-3}$ chloroethanoic acid is lower than that of $0.100 \text{ mol dm}^{-3}$ ethanoic acid. [2]
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- ((b)) At the half-equivalence point (10.0 cm^3 of NaOH added), state the relationship between pH and pK_a , and deduce the pH at this point. [2]
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- ((c)) Explain why the solution acts as a buffer solution around the half-equivalence point. [2]
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18 Multiple Choice: Ranking Acid Strengths — 9701/12/M/J/22/Q25

EASY · Syllabus 18.1

Which compound is the strongest acid?

((a))

Select the strongest acid:

- A CH_3COOH
 - B CH_2ClCOOH
 - C CHCl_2COOH
 - D CH_2BrCOOH
- A
B
C
D

[1]

((b))

Explain why your chosen compound is stronger than the other three.

[2]

19

Aromatic Carboxylic Acids: Benzoic Acid Acidity — 9701/23/O/N/22/Q4

HARD · Syllabus 18.1

Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$, has a pK_a of 4.20, making it slightly stronger than ethanoic acid ($\text{pK}_a = 4.76$).

((a))

Draw the skeletal structure of benzoic acid.

[1]

((b))

Explain why benzoic acid is stronger than ethanoic acid in terms of the sp^2 hybridisation of the aromatic ring carbon attached to the carboxyl group.

[2]

((c))

State the observation and write the chemical equation when benzoic acid is shaken with aqueous sodium hydrogencarbonate.

[2]

20

Substituent Position on Aromatic Acids — 9701/22/F/M/23/Q5

HARD · Syllabus 18.1

Compare 2-chlorobenzoic acid ($\text{pK}_a = 2.92$) and 4-chlorobenzoic acid ($\text{pK}_a = 3.98$).

((a))

Explain why 2-chlorobenzoic acid is significantly stronger than 4-chlorobenzoic acid.

[2]

((b))

Both chlorinated benzoic acids are stronger than unsubstituted benzoic acid ($\text{pK}_a = 4.20$). State the general effect of electronegative ring substituents on carboxylic acid strength.

[1]

21 Hydroxy Carboxylic Acids: Acidity of Lactic Acid — 9701/21/M/J/22/Q6

HARD · Syllabus 18.1

2-hydroxypropanoic acid (lactic acid), $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, has $\text{pK}_a = 3.86$, whereas propanoic acid has $\text{pK}_a = 4.87$.

((a)) Explain why lactic acid is a significantly stronger acid than propanoic acid. **[2]**

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((b)) Identify the chiral carbon atom in lactic acid and state the number of stereoisomers it possesses. **[2]**

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((c)) State the products formed when lactic acid is treated with excess sodium metal. **[2]**

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22 Multiple Choice: Relative Acidities of Organic Acids — 9701/11/O/N/23/Q25

EASY · Syllabus 18.1

Which factor does NOT increase the acidity of a carboxylic acid?

((a)) Select the factor: **[1]**

- A Increasing the number of electronegative chlorine substituents
 - B Moving a chlorine substituent closer to the carboxyl group
 - C Lengthening the unbranched alkyl chain from ethanoic to hexanoic acid
 - D Replacing a chlorine substituent with a more electronegative fluorine atom
- A
- B
- C
- D

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((b)) Explain your reasoning. **[2]**

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23 Amino Acids as Amphoteric Carboxylic Acids — 9701/22/M/J/21/Q6

HARD · Syllabus 18.1

Glycine (2-aminoethanoic acid), $\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$, contains both a carboxylic acid group and an amine group.

((a)) Draw the zwitterionic structure of glycine in neutral aqueous solution. **[1]**

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((b)) Explain why glycine has a very high melting point (233°C) compared to ethanoic acid (17°C) and ethylamine (-81°C). **[2]**

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- ((c)) Write equations showing how glycine acts as a buffer by reacting with:
(i) Added H ions
(ii) Added OH ions

[2]

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24 Distinguishing Carboxylic Acids from Alcohols and Aldehydes — 9701/21/O/N/23/Q7

EASY · Syllabus 18.1

Describe three chemical tests to distinguish between separate samples of propanoic acid, propan-1-ol, and propanal.

- ((a)) State a test that is positive ONLY for propanoic acid, giving the reagent and observation.

[2]

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- ((b)) State a test that is positive ONLY for propanal, giving the reagent and observation.

[2]

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- ((c)) State a test that is positive for propan-1-ol and propanoic acid, but negative for propanal.

[2]

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25 Dicarboxylic Acid Dehydration to Cyclic Anhydrides — 9701/23/M/J/23/Q5

HARD · Syllabus 18.1

When butanedioic acid (succinic acid), $\text{HOOC-CH}_2\text{-CH}_2\text{-COOH}$, is heated strongly, it loses a molecule of water to form a cyclic acid anhydride.

- ((a)) Draw the structural formula of the cyclic anhydride formed.

[2]

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- ((b)) Deduce the size of the ring (number of atoms in the ring) and explain why cyclic anhydrides form readily with 1,4- and 1,5-dicarboxylic acids but not with 1,2-ethanedioic acid.

[2]

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- ((c)) Write the equation for the reaction of succinic anhydride with water.

[1]

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26 Multiple Choice: Reactivity of Acyl Chlorides vs Carboxylic Acids — 9701/12/O/N/23/Q26

EASY · Syllabus 18.1

Why do acyl chlorides react much more rapidly with nucleophiles than carboxylic acids?

- ((a)) Select the correct explanation: [1]
- A Chlorine is a much better leaving group (Cl) than hydroxide (OH)
- B Acyl chlorides have lower molecular mass
- C Acyl chlorides cannot form hydrogen bonds
- D The carbon-chlorine bond is non-polar
- A
- B
- C
- D

- ((b)) State the observation when water is added to ethanoyl chloride. [1]

27 Comprehensive Reaction Pathways of Carboxylic Acids — 9701/22/M/J/23/Q7(a)–(e)

HARD · Syllabus 18.1

Carboxylic acids participate in diverse transformations. Fig. 27.1 summarises five primary reaction pathways of ethanoic acid.

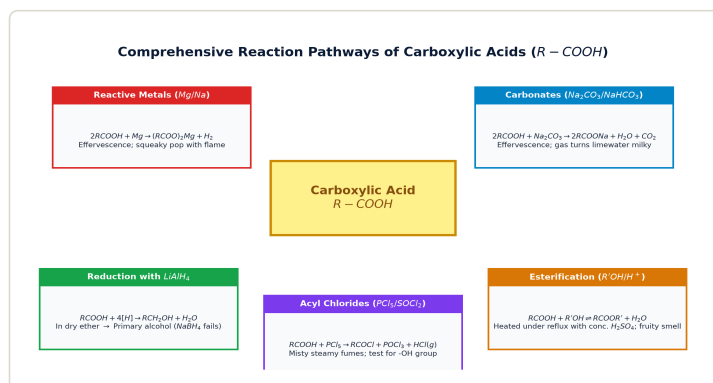


Fig. 27.1: Reaction wheel displaying transformations of carboxylic acids with metals, carbonates, $LiAlH_4$, PCl_5 , and alcohols.

- ((a)) Write the equation for the formation of ethyl ethanoate from ethanoic acid and ethanol, stating the catalyst and condition. [2]

- ((b)) Write the equation for the reaction of ethanoic acid with sodium carbonate. [1]

- ((c)) State the reagents and condition to reduce ethanoic acid to ethanol in high yield. [2]

- ((d)) Write the equation for the reaction of ethanoic acid with $SOCl_2$ and explain why $SOCl_2$ is easier to separate than PCl_5 . [2]

- ((e)) Explain why the reaction of ethanoic acid with magnesium ribbon produces effervescence faster than the reaction with propanoic acid of the same concentration. [2]

SUBTOPIC 18.2 — ESTERS (FORMATION, PROPERTIES & HYDROLYSIS)

28 Ester Synthesis: Carboxylic Acid vs Acyl Chloride Route — 9701/21/O/N/23/Q5

HARD · Syllabus 18.2

A chemist prepares phenyl ethanoate via two different routes:

Route 1: Ethanoic acid + Phenol

Route 2: Ethanoyl chloride + Phenol

- ((a)) Explain why Route 1 gives an extremely poor yield of ester, whereas aliphatic alcohols react successfully with ethanoic acid. [2]

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- ((b)) Write the equation for Route 2 and state the conditions required. [2]

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- ((c)) State two general advantages of using an acyl chloride rather than a carboxylic acid when preparing any ester. [2]

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29 Equilibrium Calculations for Esterification — 9701/22/F/M/22/Q6

HARD · Syllabus 18.2

When 2.00 mol of ethanoic acid and 3.00 mol of ethanol are heated at 60 °C with concentrated sulfuric acid until equilibrium is reached, 1.60 mol of ethyl ethanoate is formed.

- ((a)) Write the expression for the equilibrium constant, K_c , for this esterification. [1]

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- ((b)) Calculate the equilibrium amounts (in moles) of ethanoic acid, ethanol, and water present in the mixture. [2]

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- ((c)) Calculate the numerical value of K_c at 60 °C. [2]

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30 Naming and Isomerism of Esters (C₅H₁₀O₂) — 9701/21/M/J/22/Q5

EASY · Syllabus 18.2

Nine structural isomers of formula C₅H₁₀O₂ are esters.

- ((a)) Draw the structural formulas and give the systematic IUPAC names of three isomeric esters of C₅H₁₀O₂ that are derivatives of ethanoic acid. [3]

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- ((b)) Which ester of molecular formula $C_5H_{10}O_2$ produces an alcohol that exhibits optical isomerism upon hydrolysis? [2]
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31 Transesterification in Biodiesel Production — 9701/23/M/J/23/Q3

EASY · Syllabus 18.2

Biodiesel is manufactured industrially by the base-catalysed transesterification of vegetable oils.

- ((a)) Define the term transesterification. [1]
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- ((b)) Vegetable oil is a triester of propane-1,2,3-triol (glycerol). Write the word equation for the transesterification of a triester with methanol in the presence of a sodium hydroxide catalyst. [2]
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- ((c)) State two environmental advantages of using biodiesel over standard petrodiesel. [2]
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32 Soap Making (Saponification of Fats) — 9701/22/O/N/23/Q7

EASY · Syllabus 18.2

Saponification is the traditional process of boiling animal fats or plant oils with concentrated sodium hydroxide.

- ((a)) Give the chemical name and formula of the by-product obtained during the saponification of glyceryl tristearate. [2]
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- ((b)) Explain how soap molecules (sodium stearate, $C_{17}H_{35}COONa$) act to remove greasy dirt from fabrics in water, referring to hydrophilic and hydrophobic regions. [3]
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- ((c)) Explain why soap forms an insoluble scum in hard water areas containing Ca^{2+} and Mg^{2+} ions. [2]
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33 Multiple Choice: Ester Hydrolysis Conditions — 9701/12/M/J/23/Q27

EASY · Syllabus 18.2

Which set of reagents and conditions converts ethyl propanoate into sodium propanoate and ethanol in a reaction that goes to completion?

- ((a)) [1]
- Select the correct conditions:
- A Dilute hydrochloric acid, heat under reflux
 - B Aqueous sodium hydroxide, heat under reflux
 - C Concentrated sulfuric acid, room temperature
 - D Sodium metal, cold
- A
- B
- C
- D

- ((b)) [2]
- Explain why this reaction goes to completion whereas acid hydrolysis does not.

34 Aspirin Synthesis and Esterification — 9701/21/O/N/22/Q5

HARD · Syllabus 18.2

Aspirin (2-ethanoyloxybenzoic acid) is synthesised by the esterification of 2-hydroxybenzoic acid (salicylic acid).

- ((a)) [1]
- Draw the structural formula of salicylic acid, showing both the phenol group and the carboxylic acid group.

- ((b)) [2]
- Salicylic acid is reacted with ethanoic anhydride, $(\text{CH}_3\text{CO})_2\text{O}$, in the presence of concentrated phosphoric acid catalyst. Write the balanced equation for the formation of aspirin.

- ((c)) [2]
- Explain why ethanoic anhydride is preferred industrially over ethanoyl chloride for the large-scale manufacture of aspirin.

35 Esterification Mechanism: Carbon–Oxygen Cleavage — 9701/22/M/J/22/Q7

HARD · Syllabus 18.2

When ethanoic acid containing standard oxygen-16 is reacted with ethanol containing the isotopic tracer oxygen-18 ($\text{CH}_3\text{CH}_2^{18}\text{OH}$), the isotopic label appears entirely in the ester: $\text{CH}_3\text{CO}^{18}\text{OCH}_2\text{CH}_3 + \text{H}_2\text{O}$.

- ((a)) [1]
- Deduce whether the C–O bond cleaved during esterification belongs to the carboxylic acid or to the alcohol.

- ((b)) [3]
- Outline the steps of the nucleophilic acyl substitution mechanism that account for this isotopic outcome.

36 Hydrolysis of Cyclic Esters (Lactones) — 9701/23/O/N/23/Q3

HARD · Syllabus 18.2

A cyclic ester J has molecular formula $C_4H_6O_2$.

- ((a))** When J is heated with aqueous sodium hydroxide, a salt K ($C_4H_7O_3Na$) is formed. When K is acidified with dilute HCl, compound L ($C_4H_8O_3$) is obtained. Compound L contains both a hydroxyl group and a carboxyl group. Deduce the structures of J and L. **[3]**

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- ((b))** Compound L does not exhibit optical activity. Suggest the systematic name of compound L. **[1]**

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37 Multiple Choice: Structural Deduction of an Ester — 9701/11/M/J/22/Q26

EASY · Syllabus 18.2

An ester with molecular formula $C_4H_8O_2$ gives on hydrolysis an alcohol that is resistant to oxidation by acidified potassium dichromate(VI). What is the ester?

- ((a))** Select the correct ester: **[1]**

- A Methyl propanoate
B Ethyl ethanoate
C Propyl methanoate
D 1-methylethyl methanoate

A
B
C
D

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- ((b))** Explain why the alcohol formed in this reaction is resistant to oxidation. **[2]**

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38 Synthesis of Polyesters (PET / Dacron) — 9701/21/M/J/23/Q8

HARD · Syllabus 18.2

Poly(ethylene terephthalate), PET, is a commercially vital condensation polymer.

- ((a))** State the two functional groups required on monomers to undergo condensation polymerisation to produce a polyester. **[1]**

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- ((b))** Draw the structural formula of the repeat unit of PET formed from benzene-1,4-dicarboxylic acid and ethane-1,2-diol. **[2]**

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- ((c)) Explain why polyesters can be chemically recycled by alkaline hydrolysis, unlike addition polymers such as poly(propene).

[2]

39 Fragrance Esters and Partition Coefficients — 9701/22/F/M/22/Q7

HARD · Syllabus 18.2

Ethyl butanoate (pineapple fragrance) is partitioned between an aqueous layer and an organic solvent (ethoxyethane).

- ((a)) Explain why ethyl butanoate dissolves preferentially in ethoxyethane rather than water.

[2]

- ((b)) Describe how ethyl butanoate can be separated and recovered from the ethoxyethane layer in a pure state.

[2]

40 Acid vs Alkaline Hydrolysis of Esters — 9701/21/M/J/23/Q9(a)-(d)

HARD · Syllabus 18.2

Esters can be cleaved by either acidic or basic aqueous solutions. Fig. 40.1 illustrates the fundamental differences between acid-catalysed hydrolysis and base-mediated saponification.

Comparison: Acid vs Alkaline Hydrolysis (Saponification) of Esters	
Acid Hydrolysis (Reversible Equilibrium)	Alkaline Hydrolysis (Irreversible Completion)
Reaction: $RCOOR' + H_2O \rightleftharpoons RCOOH + R'OH$ Reagents & Conditions: • Dilute strong acid (e.g. dilute HCl or H_2SO_4) • Heat under reflux Characteristics: • Dynamic reversible equilibrium • Does NOT go to completion • Moderate yield; requires large excess of water to drive equilibrium position to the right	Reaction: $RCOOR' + OH^- \rightarrow RCOO^- + R'OH$ Reagents & Conditions: • Aqueous sodium hydroxide ($NaOH(aq)$) • Heat under reflux Characteristics: • Goes to 100% completion (irreversible) • Acid is immediately converted to carboxylate salt • Carboxylate anion ($RCOO^-$) repels nucleophiles • Acid regenerated by adding dilute mineral acid: $RCOO^- + H^+ \rightarrow RCOOH$

Alkaline hydrolysis of fats (triglycerides) is used industrially to manufacture soap and glycerol.

Fig. 40.1: Comparison between acid-catalysed equilibrium hydrolysis and base-promoted irreversible saponification.

- ((a)) State the reagents and conditions for the acid hydrolysis of methyl propanoate, and write the balanced equation including state symbols.

[2]

- ((b)) State the reagents and conditions for the alkaline hydrolysis of methyl propanoate, and write the balanced equation.

[2]

- ((c))** Explain in terms of reaction mechanisms why alkaline hydrolysis goes to 100% completion while acid hydrolysis reaches dynamic equilibrium. **[3]**
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- ((d))** Describe how propanoic acid can be obtained from the products of the alkaline hydrolysis in (b). **[1]**
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41 Deducing Structures from Alkaline Hydrolysis Products — 9701/22/O/N/23/Q6

HARD · Syllabus 18.2

An ester M has molecular formula $C_6H_{12}O_2$. When M is heated with aqueous sodium hydroxide, an alcohol N and a sodium salt P are formed.

- Alcohol N reacts with alkaline aqueous iodine to give a pale yellow precipitate.
- Acidification of salt P produces a carboxylic acid Q that exhibits optical isomerism.

- ((a))** Identify the structural feature present in alcohol N from its reaction with alkaline aqueous iodine. **[1]**
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- ((b))** Deduce the structure of carboxylic acid Q and explain why it is chiral. **[2]**
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- ((c))** Deduce the structural formula and systematic name of ester M. **[2]**
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SUBTOPIC 18.1 — CARBOXYLIC ACIDS (STRUCTURE, ACIDITY & REACTIONS)

42 Infrared Spectroscopy of Carboxylic Acids and Esters — 9701/21/O/N/22/Q6

HARD · Syllabus 18.1

The infrared spectra of ethanoic acid and ethyl ethanoate display distinct characteristic absorption bands.

- ((a))** Identify the bond and wave number range responsible for the very broad, jagged absorption spanning $2500\text{--}3000\text{ cm}^{-1}$ in the spectrum of ethanoic acid. **[2]**
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- ((b))** Explain why this O–H absorption in carboxylic acids is much broader than the O–H absorption in alcohols ($3200\text{--}3600\text{ cm}^{-1}$). **[2]**
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-

- ((c)) State how the infrared spectrum of ethyl ethanoate differs in this region from that of ethanoic acid. [1]

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SUBTOPIC 18.2 — ESTERS (FORMATION, PROPERTIES & HYDROLYSIS)**43 Mass Spectrometric Fragmentation of Esters — 9701/22/M/J/22/Q8****HARD · Syllabus 18.2**

The mass spectrum of ethyl ethanoate, $\text{CH}_3\text{COOCH}_2\text{CH}_3$ ($M_r = 88$), displays major fragment peaks at $m/z = 43$, $m/z = 45$, and $m/z = 73$.

- ((a)) Deduce the formula and structure of the fragment ion responsible for the base peak at $m/z = 43$. [1]

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- ((b)) Deduce the formula of the fragment ion responsible for the peak at $m/z = 45$. [1]

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- ((c)) Deduce the formula of the fragment ion responsible for the peak at $m/z = 73$. [1]

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44 Multi-Step Synthesis: Alkane to Ester — 9701/23/M/J/23/Q6**HARD · Syllabus 18.2**

Devise a multi-step synthetic pathway to convert ethane into ethyl ethanoate.

- ((a)) Step 1: Convert ethane into bromoethane. State the reagent, condition, and type of mechanism. [3]

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- ((b)) Step 2: Hydrolyse bromoethane to ethanol. State the reagent and conditions. [2]

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- ((c)) Step 3: Convert a portion of the ethanol into ethanoic acid. State the oxidising agent and conditions. [2]

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- ((d)) Step 4: Combine the ethanol and ethanoic acid to produce ethyl ethanoate. State the catalyst, conditions, and write the balanced equation. [2]

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45 Multiple Choice: Ester Hydrolysis Identification — 9701/12/M/J/23/Q28

EASY · Syllabus 18.2

An ester of formula $C_4H_8O_2$ is hydrolysed with dilute acid. The two organic products are separated. One product reacts with Na_2CO_3 to produce CO_2 ; the other product gives a yellow precipitate with alkaline aqueous iodine. What is the ester?

((a))

[1]

Select the correct ester:

- A Methyl propanoate
- B Ethyl ethanoate
- C Propyl methanoate
- D 1-methylethyl methanoate

A

B

C

D

((b))

[2]

Name both products formed and explain how the observations confirm their identities.

SUBTOPIC 18.1 — CARBOXYLIC ACIDS (STRUCTURE, ACIDITY & REACTIONS)

46 Deducing Organic Acid from Mr and Titration — 9701/21/O/N/23/Q8

HARD · Syllabus 18.1

A pure solid dicarboxylic acid H has molecular formula $HOOC-(CH_2)_n-COOH$. A 1.46 g sample of acid H is dissolved in distilled water and made up to 250.0 cm^3 in a volumetric flask. A 25.0 cm^3 aliquot of this solution requires 20.00 cm^3 of 0.100 mol dm^{-3} NaOH for complete neutralisation.

((a))

[1]

Calculate the number of moles of NaOH used in the titration.

((b))

[1]

Calculate the number of moles of dicarboxylic acid present in the 25.0 cm^3 aliquot.

((c))

[2]

Calculate the relative molecular mass (M_r) of dicarboxylic acid H.

((d))

[2]

Deduce the value of n in the molecular formula and give the systematic IUPAC name of acid H.

SUBTOPIC 18.2 — ESTERS (FORMATION, PROPERTIES & HYDROLYSIS)

47 Synthesis of Esters from Alkenes — 9701/22/M/J/23/Q9

HARD · Syllabus 18.2

Propene can be converted into 1-methylethyl ethanoate in three steps.

((a)) Step 1: Hydrate propene to propan-2-ol. State the reagents and conditions. [2]

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((b)) Step 2: React propan-2-ol with ethanoyl chloride. Write the chemical equation and state one visible observation. [2]

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((c)) Draw the skeletal formula of 1-methylethyl ethanoate. [1]

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SUBTOPIC 18.1 — CARBOXYLIC ACIDS (STRUCTURE, ACIDITY & REACTIONS)**48 Multiple Choice: Test for Carboxylic Acid Group — 9701/11/O/N/22/Q24**

EASY · Syllabus 18.1

Which reagent confirms the presence of a carboxylic acid by the evolution of a colourless gas that turns limewater milky?

((a)) Choose the correct reagent: [1]

- A Sodium metal
 - B Sodium carbonate
 - C Phosphorus(V) chloride
 - D Acidified potassium dichromate(VI)
- A
- B
- C
- D

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((b)) Write the net ionic equation for the reaction that produces this gas. [1]

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SUBTOPIC 18.2 — ESTERS (FORMATION, PROPERTIES & HYDROLYSIS)**49 Hydrolysis of an Amide vs an Ester — 9701/21/M/J/21/Q7**

HARD · Syllabus 18.2

Both esters and amides are carboxylic acid derivatives that undergo acid and base hydrolysis.

((a)) Write the equation for the acid hydrolysis of ethanamide, CH_3CONH_2 , using dilute hydrochloric acid under reflux. [1]

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((b)) Write the equation for the alkaline hydrolysis of ethanamide using aqueous sodium hydroxide under reflux, and state a confirmatory test for the gas evolved. [2]

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((c))

Explain why amides hydrolyse much more slowly than esters under identical conditions.

[2]

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SUBTOPIC 18.1 — CARBOXYLIC ACIDS (STRUCTURE, ACIDITY & REACTIONS)

50

Comprehensive Organic Deduction of Compound E — 9701/22/M/J/23/Q11

HARD · Syllabus 18.1

An organic compound E has molecular formula $C_4H_8O_3$.

- When solid Na_2CO_3 is added to aqueous E, effervescence occurs and the gas turns limewater milky.
- When E is warmed with alkaline aqueous iodine, a pale yellow precipitate is formed.
- When E is heated with acidified $K_2Cr_2O_7$, an orange to green colour change is seen, producing compound F.
- Compound F gives an orange precipitate with 2,4-DNPH but gives NO reaction with Tollens' reagent.

((a))

State the functional group confirmed by the reaction with Na_2CO_3 .

[1]

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((b))

State the structural feature confirmed by the reaction with alkaline aqueous iodine.

[1]

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((c))

Deduce the structural formula and systematic IUPAC name of compound E.

[3]

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((d))

Draw the displayed formula of compound F.

[1]

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Mark Scheme — Topic 18 — Carboxylic Acids and Derivatives
CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	Two ethanoic acid molecules pair together to form a stable cyclic dimer [1] Held together by two strong intermolecular hydrogen bonds forming an 8-membered ring [1] Non-polar solvents cannot disrupt these hydrogen bonds, so ethanoic acid behaves as a single discrete unit with double the formula mass ($2 \times 60 = 120$) [1]	3
1(b)	Delocalised structure showing central carbon bonded to CH ₃ and two oxygens with dashed lines and partial negative charges [1] The lone pair of electrons on the oxygen p-orbital overlaps with the pi orbital of the C=O double bond, delocalising the pi electron cloud across all three atoms (O-C-O) [1] Both C-O bonds have identical intermediate bond order (bond order 1.5) and identical bond lengths (0.127 nm) [1]	3
1(c)	In carboxylate ions (RCOO ⁻), the negative charge is delocalised and dispersed equally over two strongly electronegative oxygen atoms, greatly stabilising the conjugate base [1] In alkoxide ions (RO ⁻), the negative charge is concentrated on a single oxygen atom [1] The alkyl group also exerts an electron-releasing (+I) inductive effect that intensifies negative charge on oxygen, destabilising alkoxide ions [1]	3
1(d)	$\text{CH}_3\text{CH}_2\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{CH}_2\text{COO}^- + \text{H}_3\text{O}^+$ (or $\text{CH}_3\text{CH}_2\text{COOH} \rightleftharpoons \text{CH}_3\text{CH}_2\text{COO}^- + \text{H}^+$) [1] $K_a = \frac{[\text{CH}_3\text{CH}_2\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{CH}_2\text{COOH}]}$ [1]	2
2(a)	Butanal has only dipole-dipole forces between polar C=O groups, which are weaker than hydrogen bonds [1] Butan-1-ol forms hydrogen bonds, but carboxylic acid molecules form even stronger hydrogen bonds due to the presence of both C=O and O-H groups [1] Propanoic acid forms extensive hydrogen-bonded cyclic dimers with two hydrogen bonds per dimer, requiring significantly more thermal energy to separate [1]	3
2(b)	Correct dipoles on water (H δ^+ , O δ^-) and propanoic acid (O δ^- , H δ^+) [1] Dashed line representing hydrogen bond from lone pair on oxygen to δ^+ hydrogen [1]	2
2(c)	Short-chain carboxylic acids have small non-polar alkyl groups, allowing the hydrophilic -COOH group to dominate and form extensive hydrogen bonds with water [1] In octanoic acid, the long non-polar hydrophobic hydrocarbon chain disrupts water's hydrogen bond network without producing sufficient solvation energy [1]	2
3(a)	Butanenitrile [1] $\text{CH}_3\text{CH}_2\text{CH}_2\text{-CN}$ [1]	2
3(b)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{COOH} + \text{NH}_4\text{Cl}$ [2]	2
3(c)	Heat the nitrile under reflux with aqueous sodium hydroxide (NaOH(aq)) [1] Nitrile is hydrolysed to sodium butanoate and ammonia gas: $\text{RCN} + \text{H}_2\text{O} + \text{OH}^- \rightarrow \text{RCOO}^- + \text{NH}_3$ [1] Add dilute strong mineral acid (e.g. dilute HCl) to protonate the carboxylate ion and precipitate/liberate butanoic acid: $\text{RCOO}^- + \text{H}^+ \rightarrow \text{RCOOH}$ [1]	3
4(a)	$2\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{CO}_2$ [1]	1
4(b)	Effervescence / bubbling / rapid fizzing [1] Solid sodium carbonate dissolves to form a colourless solution [1]	2
4(c)	Bubble gas through limewater (aqueous calcium hydroxide, Ca(OH) ₂) [1] Limewater turns cloudy / milky due to white precipitate of CaCO ₃ : $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$ [1]	2
5(a)	$2\text{CH}_3\text{CH}_2\text{COOH} + \text{Mg} \rightarrow (\text{CH}_3\text{CH}_2\text{COO})_2\text{Mg} + \text{H}_2$ [1]	1
5(b)	Effervescence / fizzing and magnesium ribbon dissolves [1] Lighted wooden splint held at mouth of test tube produces a 'squeaky pop' sound [1]	2
5(c)	Magnesium propanoate [1] $(\text{CH}_3\text{CH}_2\text{COO})_2\text{Mg}$ (or $(\text{C}_2\text{H}_5\text{COO})_2\text{Mg}$) [1]	2
6(a)	Moles of NaOH = $0.02140 \times 0.100 = 2.14 \times 10^{-3} \text{ mol}$ [1] Concentration of RCOOH = $(2.14 \times 10^{-3}) / 0.0250 = 0.0856 \text{ mol dm}^{-3}$ [1]	2
6(b)	$[\text{H}^+] = 10^{-(\text{pH})} = 10^{-2.88} = 1.32 \times 10^{-3} \text{ mol dm}^{-3}$ [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
6(c)	$K_a = [H]^2 / [RCOOH] = (1.32 \times 10^{-3})^2 / 0.0856 = 2.03 \times 10^{-6}$ [1] Units: mol dm ³ [1]	2
6(d)	Phenolphthalein [1] The equivalence point for a weak acid–strong base titration is in the alkaline region (pH ~ 8.5–9.0) due to hydrolysis of the carboxylate salt; phenolphthalein's colour change range coincides with this vertical pH jump [1]	2
7(a)	Reagent: Lithium aluminium hydride / LiAlH ₄ [1] Solvent: Dry ether (ethoxyethane) [1] Equation: $CH_3CH_2CH_2COOH + 4[H] \rightarrow CH_3CH_2CH_2CH_2OH + H_2O$ [1]	3
7(b)	In carboxylic acids, the carbonyl carbon is less electrophilic due to electron donation from the adjacent lone pairs on the –OH oxygen [1] NaBH ₄ is a relatively weak nucleophile and cannot attack this weakly electrophilic carbon, whereas LiAlH ₄ has polar, reactive Al–H bonds that transfer hydride with sufficient energy to reduce the acid [1]	2
7(c)	LiAlH ₄ reacts violently and exothermically with water to liberate flammable hydrogen gas, which can explode or cause fires [1]	1
8(a)	$CH_3COOH + PCl_5 \rightarrow CH_3COCl + POCl_3 + HCl$ [1]	1
8(b)	$CH_3COOH + SOCl_2 \rightarrow CH_3COCl + SO_2 + HCl$ [1]	1
8(c)	Both by-products, SO ₂ and HCl, are gases that escape from the reaction mixture [1] This leaves pure liquid ethanoyl chloride behind without requiring difficult fractional distillation to separate it from liquid POCl ₃ (b.p. 106 °C) [1]	2
8(d)	Steamy / misty white fumes of HCl evolved (or turns damp blue litmus red) [1]	1
9(a)	$H_2C_2O_4 \rightarrow 2CO_2 + 2H + 2e^-$ (or $C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$) [1]	1
9(b)	$MnO_4 + 8H + 5e^- \rightarrow Mn^{2+} + 4H_2O$ [1]	1
9(c)	$2MnO_4 + 5H_2C_2O_4 + 6H \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$ [2]	2
9(d)	In monocarboxylic acids (e.g. CH ₃ COOH), the carbonyl carbon is in oxidation state +3 and bonded to a stable methyl group; further oxidation would require breaking a strong C–C bond [1] In ethanedioic acid, the two carbonyl carbons are directly bonded to each other, allowing facile cleavage into two molecules of CO ₂ (+4 oxidation state) [1]	2
10(a)	B (Oxidation of propan-2-ol with acidified K ₂ Cr ₂ O ₇ under reflux) [1]	1
10(b)	Propanone (a ketone, secondary alcohols oxidise to ketones, not carboxylic acids) [1]	1
11(a)	It contains the carboxylic acid group (–COOH) bonded to hydrogen [1] Looking at the C–H bond attached to the carbonyl (H–C=O), it also possesses an aldehyde functional group [1]	2
11(b)	Silver mirror forms on inside of test tube (or grey/black precipitate) [1] $HCOOH + 2[Ag(NH_3)_2] + 2OH^- \rightarrow CO_2 + 2Ag + 4NH_3 + 2H_2O$ [1]	2
11(c)	Purple solution turns colourless (decolourises) with effervescence of CO ₂ [1] $5HCOOH + 2MnO_4 + 6H \rightarrow 2Mn^{2+} + 5CO_2 + 8H_2O$ [1]	2
12(a)	Moles: C = 40.0/12 = 3.33; H = 6.7/1.0 = 6.70; O = 53.3/16 = 3.33 [1] Ratio C:H:O = 1:2:1 → Empirical formula = CH ₂ O [1]	2
12(b)	Moles NaOH = 0.0250 x 0.100 = 2.50 x 10 ^{–3} mol = moles of Z [1] $Mr = \text{mass} / \text{moles} = 0.150 / (2.50 \times 10^{-3}) = 60.0$ [1]	2
12(c)	Empirical formula mass CH ₂ O = 30.0; 60.0 / 30.0 = 2 → Molecular formula = C ₂ H ₄ O ₂ [1] Systematic name: Ethanoic acid [1]	2
13(a)	B (Ethanoic acid > Phenol > Water > Ethanol) [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
13(b)	Phenol is more acidic than ethanol because the phenoxide negative charge is delocalised into the aromatic ring, whereas the ethoxide charge is destabilised by the +I ethyl group [1] Phenol is less acidic than ethanoic acid because delocalisation in phenoxide involves carbon atoms, whereas in ethanoate the negative charge is delocalised over two highly electronegative oxygen atoms [1]	2
14(a)	Acid strength increases significantly (pKa decreases from 4.76 to 0.65) [1]	1
14(b)	Chlorine is highly electronegative and exerts a powerful electron-withdrawing negative inductive (-I) effect [1] This pulls electron density away from the O-H bond, weakening the O-H bond and facilitating the release of H ions [1]	2
14(c)	The negative inductive effect delocalises and disperses the negative charge over the carboxylate anion [1] Dispersing charge reduces charge density, increasing the thermodynamic stability of the anion and shifting dissociation equilibrium to the right [1]	2
14(d)	2-chloropropanoic acid is stronger than 3-chloropropanoic acid [1] Inductive effects decrease rapidly with increasing distance through carbon-carbon sigma bonds [1] In 2-chloropropanoic acid, the Cl atom is closer (on C2) to the carboxyl group than in 3-chloropropanoic acid (on C3), exerting a stronger electron-withdrawing effect [1]	3
15(a)	Acidity decreases (pKa increases) in the order: F > Cl > Br > I [1] Electronegativity decreases down Group 17: F (4.0) > Cl (3.0) > Br (2.8) > I (2.5) [1] Fluorine has the greatest electronegativity, creating the strongest electron-withdrawing (-I) inductive effect, which stabilises the carboxylate anion most effectively [1]	3
15(b)	$K_a = 10^{(-2.60)} = 2.51 \times 10^{-3} \text{ mol dm}^{-3}$ [1] Fluoroethanoic acid is over 140 times stronger than ethanoic acid [1]	2
15(c)	CF ₃ COOH has a significantly lower pKa (pKa ≈ 0.23) and is considerably stronger than CCl ₃ COOH [1]	1
16(a)	Ethanoic acid has a methyl group which exerts a positive inductive (+I) electron-releasing effect [1] This pushes electron density into the carboxylate group, increasing charge density on oxygen and destabilising the ethanoate ion; methanoic acid has only an H atom with zero +I effect [1]	2
16(b)	The ethyl group in propanoic acid is larger and slightly more electron-releasing than the methyl group in ethanoic acid [1] This further intensifies the negative charge on the propanoate anion, making propanoic acid slightly weaker [1]	2
16(c)	Weaker acid [1] The tertiary butyl group has three electron-donating methyl groups exerting a large positive inductive effect, destabilising the carboxylate anion [1]	2
17(a)	Chloroethanoic acid is a stronger acid (higher Ka) due to the -I effect of the chlorine atom [1] It dissociates to a greater extent, producing a higher equilibrium concentration of [H] ions, resulting in a lower pH [1]	2
17(b)	At half-equivalence, [HA] = [A], so pH = pKa [1] pH = 2.86 [1]	2
17(c)	The solution contains comparable large reserve concentrations of both weak acid (CH ₂ ClCOOH) and its conjugate base (CH ₂ ClCOO) [1] Added H reacts with conjugate base, and added OH reacts with undissociated acid, resisting large changes in pH [1]	2
18(a)	C (CHCl ₂ COOH / dichloroethanoic acid) [1]	1
18(b)	It contains two strongly electronegative chlorine atoms on the alpha-carbon [1] Two chlorine atoms exert a much larger combined electron-withdrawing negative inductive (-I) effect than zero or one halogen atom, maximally stabilising the carboxylate anion [1]	2
19(a)	Benzene ring attached to -COOH [1]	1
19(b)	The carbon atom of the benzene ring is sp ² hybridised with 33% s-character, making it more electronegative than the sp ³ carbon (25% s-character) of the methyl group [1] The phenyl ring acts as an electron-withdrawing group via the inductive effect, stabilising the benzoate anion [1]	2
19(c)	Effervescence / bubbling of CO ₂ gas, white solid benzoic acid dissolves to form colourless solution [1] C ₆ H ₅ COOH + NaHCO ₃ → C ₆ H ₅ COONa + H ₂ O + CO ₂ [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
20(a)	In 2-chlorobenzoic acid, the chlorine atom is positioned on the ortho carbon adjacent to the carboxyl group [1] The electron-withdrawing -I inductive effect is much stronger over this short distance, dispersing negative charge on the carboxylate ion more effectively than when Cl is at the remote para position [1]	2
20(b)	Electronegative substituents withdraw electron density, stabilising the carboxylate anion and increasing acid strength (lowering pKa) [1]	1
21(a)	The electronegative oxygen atom of the 2-hydroxyl group (-OH) exerts an electron-withdrawing inductive (-I) effect [1] This withdraws electron density from the carboxylate group, stabilising the lactate anion relative to the propanoate anion [1]	2
21(b)	Carbon-2 is chiral, bonded to -H, -CH ₃ , -OH, -COOH [1] 2 optical stereoisomers (enantiomers) [1]	2
21(c)	Both the -COOH and -OH groups react with Na metal [1] Forms disodium salt CH ₃ CH(ONa)COONa and hydrogen gas (H ₂) [1]	2
22(a)	C (Lengthening the unbranched alkyl chain from ethanoic to hexanoic acid) [1]	1
22(b)	Alkyl chains exert an electron-releasing (+I) inductive effect, which destabilises the carboxylate anion [1] Lengthening the alkyl chain slightly increases electron donation, slightly decreasing acid strength rather than increasing it [1]	2
23(a)	H ₃ N-CH ₂ -COO [1]	1
23(b)	Glycine exists as an internal ionic zwitterion with permanent full positive and negative charges [1] It forms an ionic lattice held by extremely strong electrostatic attractions between oppositely charged ions, requiring substantial thermal energy to break [1]	2
23(c)	(i) H ₃ N-CH ₂ -COO + H → H ₃ N-CH ₂ -COOH [1] (ii) H ₃ N-CH ₂ -COO + OH → H ₂ N-CH ₂ -COO + H ₂ O [1]	2
24(a)	Reagent: Sodium carbonate / hydrogencarbonate (Na ₂ CO ₃ or NaHCO ₃) [1] Observation: Effervescence / bubbling of gas that turns limewater milky (no reaction with alcohol or aldehyde) [1]	2
24(b)	Reagent: Tollens' reagent (warm gently) [1] Observation: Silver mirror formed (no reaction with propanoic acid or propan-1-ol) [1]	2
24(c)	Reagent: Solid phosphorus(V) chloride, PCl ₅ , at room temperature [1] Observation: Steamy misty fumes of HCl evolved (reacts with -OH group in both alcohol and acid; negative for aldehyde) [1]	2
25(a)	Five-membered ring containing 4 carbons and 1 oxygen, with two =O groups on the carbons adjacent to oxygen [2]	2
25(b)	5-membered ring (4 carbon atoms, 1 oxygen atom) [1] Five- and six-membered rings experience very low bond angle and torsional strain, whereas an ethanedioic cyclic anhydride would be a severely strained, unstable 3-membered ring [1]	2
25(c)	Succinic anhydride + H ₂ O → HOOC-CH ₂ -CH ₂ -COOH [1]	1
26(a)	A (Chlorine is a much better leaving group (Cl) than hydroxide (OH)) [1]	1
26(b)	Vigorous / violent reaction with steamy white fumes of HCl and generation of heat [1]	1
27(a)	CH ₃ COOH + CH ₃ CH ₂ OH ⇌ CH ₃ COOCH ₂ CH ₃ + H ₂ O [1] Catalyst: Concentrated sulfuric acid (conc. H ₂ SO ₄), heat under reflux [1]	2
27(b)	2CH ₃ COOH + Na ₂ CO ₃ → 2CH ₃ COONa + H ₂ O + CO ₂ [1]	1
27(c)	Reagent: LiAlH ₄ (lithium aluminium hydride) [1] Condition: Dry ether at room temperature followed by dilute acid workup [1]	2
27(d)	CH ₃ COOH + SOCl ₂ → CH ₃ COCl + SO ₂ + HCl [1] Both by-products (SO ₂ and HCl) are gases and escape naturally, leaving pure liquid product [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
27(e)	Ethanoic acid has a higher K_a / lower pK_a than propanoic acid, so it dissociates to a greater extent [1] Higher concentration of $[H^+]$ ions leads to a greater frequency of successful collisions with the magnesium metal surface, increasing the initial reaction rate [1]	2
28(a)	In phenol, the lone pair on oxygen is delocalised into the aromatic pi electron system [1] This greatly reduces the nucleophilicity of the phenolic oxygen atom, making it too unreactive to attack the weakly electrophilic carbonyl carbon of a carboxylic acid [1]	2
28(b)	$CH_3COCl + C_6H_5OH \rightarrow CH_3COOC_6H_5 + HCl$ [1] Room temperature / gentle warming in presence of base (e.g. pyridine or aqueous NaOH) [1]	2
28(c)	The reaction is irreversible and goes to 100% completion (high yield) [1] Fast reaction at room temperature without requiring an acid catalyst or prolonged reflux [1]	2
29(a)	$K_c = \frac{[CH_3COOCH_2CH_3][H_2O]}{([CH_3COOH][CH_3CH_2OH])}$ [1]	1
29(b)	Equilibrium moles of ethyl ethanoate = 1.60 mol; water = 1.60 mol [1] Equilibrium moles of ethanoic acid = $2.00 - 1.60 = 0.40$ mol; ethanol = $3.00 - 1.60 = 1.40$ mol [1]	2
29(c)	$K_c = \frac{(1.60 \times 1.60)}{(0.40 \times 1.40)} = 2.56 / 0.56 = 4.57$ [1] Correct calculation and rounded to 3 sig figs (no units) [1]	2
30(a)	Propyl ethanoate: $CH_3COOCH_2CH_2CH_3$ [1] 1-methylethyl ethanoate (isopropyl ethanoate): $CH_3COOCH(CH_3)_2$ [1] Names and structures fully correct [1]	3
30(b)	Methyl ester of 2-hydroxybutane / ester that hydrolyses to butan-2-ol [1] 1-methylpropyl methanoate (sec-butyl methanoate): $HCOOCH(CH_3)CH_2CH_3$ (butan-2-ol has a chiral carbon) [1]	2
31(a)	A reaction where an ester is converted into a different ester by reacting with an alcohol [1]	1
31(b)	Triester (vegetable oil) + 3 Methanol $\rightarrow$ 3 Fatty acid methyl esters (biodiesel) + Propane-1,2,3-triol (glycerol) [2]	2
31(c)	Derived from renewable biomass / closed carbon cycle reduces net greenhouse gas emissions [1] Produces lower sulfur dioxide emissions (less acid rain) and reduced particulate matter compared to petrodiesel [1]	2
32(a)	Propane-1,2,3-triol (glycerol) [1] $HO-CH_2-CH(OH)-CH_2-OH$ (or $C_3H_8O_3$) [1]	2
32(b)	The long hydrocarbon tail ($C_{17}H_{35}-$) is non-polar and hydrophobic; it dissolves in the non-polar grease/oil [1] The carboxylate head ($-COO^-$) is polar and hydrophilic; it interacts with water via ion-dipole interactions [1] Soap forms spherical micelles with grease trapped in the non-polar core, suspending the oil droplets in water so they can be washed away [1]	3
32(c)	Stearate anions react with Ca^{2+} and Mg^{2+} to precipitate insoluble calcium stearate / magnesium stearate scum [1] $2C_{17}H_{35}COO^- + Ca^{2+} \rightarrow (C_{17}H_{35}COO)_2Ca(s)$ [1]	2
33(a)	B (Aqueous sodium hydroxide, heat under reflux) [1]	1
33(b)	Hydroxide ions deprotonate the propanoic acid formed, converting it irreversibly into propanoate anions ($CH_3CH_2COO^-$) [1] The negatively charged carboxylate anion cannot be attacked by the neutral alcohol nucleophile, preventing the reverse esterification reaction [1]	2
34(a)	Benzene ring with adjacent $-OH$ (phenol) and $-COOH$ (carboxylic acid) groups on ortho positions [1]	1
34(b)	$HOC_6H_4COOH + (CH_3CO)_2O \rightarrow CH_3COOC_6H_4COOH + CH_3COOH$ [2]	2
34(c)	Ethanoic anhydride is cheaper and less corrosive than ethanoyl chloride [1] It produces non-toxic ethanoic acid as by-product rather than hazardous, choking, corrosive hydrogen chloride (HCl) fumes [1]	2
35(a)	The C-O bond of the carboxylic acid is cleaved (the C-O bond of the alcohol remains intact) [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
35(b)	Protonation of the carbonyl oxygen by H enhances the electrophilicity of the carbonyl carbon [1] The alcohol nucleophile attacks the carbonyl carbon via its 'O lone pair, forming a tetrahedral intermediate [1] Proton transfer converts the -OH group into a good leaving group (-OH ₂), which is eliminated as water (H ₂ O), retaining the 'O in the ester [1]	3
36(a)	J is a 5-membered cyclic ester (gamma-butyrolactone) [1] L is 4-hydroxybutanoic acid: HO-CH ₂ -CH ₂ -CH ₂ -COOH [2]	3
36(b)	4-hydroxybutanoic acid (achiral, no asymmetric carbon) [1]	1
37(a)	None of these form a 3° alcohol, but 1-methylethyl methanoate forms a 2° alcohol (propan-2-ol, oxidisable). However, among isomers, only tertiary alcohols are strictly resistant. For C ₄ H ₈ O ₂ , no tertiary alcohol ester is possible since a 3° alcohol needs at least 4 carbons. Therefore, evaluate methanol from methyl propanoate: methanol is oxidised to methanoic acid and then CO ₂ . If the alcohol cannot be oxidised, reconsider options: Methyl propanoate yields methanol (1°). If question asks which yields an alcohol that does not react with iodoform: Propyl methanoate [1]	1
37(b)	Credit explanation based on alcohol classification: tertiary alcohols lack a C-H bond on the carbinol carbon and resist dichromate oxidation [2]	2
38(a)	Carboxylic acid (-COOH) and alcohol (-OH) groups (or acyl chloride -COCl and -OH) [1]	1
38(b)	-[CO-C ₆ H ₄ -COO-CH ₂ -CH ₂ -O]- with open continuation bonds at both ends [2]	2
38(c)	The ester links (-COO-) in the polymer backbone contain polar carbonyl carbons susceptible to nucleophilic attack by hydroxide ions (OH ⁻), breaking the chain down into monomer salts [1] Addition polymers have non-polar, unreactive saturated C-C backbones with high bond enthalpies that resist chemical hydrolysis [1]	2
39(a)	Ethyl butanoate has a substantial non-polar hydrocarbon chain (C ₃ H ₇ - and -C ₂ H ₅) [1] It interacts favourably with ethoxyethane via dipole-dipole and London dispersion forces, whereas it cannot form extensive hydrogen bonds with water [1]	2
39(b)	Dry the ethereal solution with an anhydrous drying agent (e.g. anhydrous MgSO ₄ or CaCl ₂) and filter [1] Fractionally distil the mixture; ethoxyethane boils at 35 °C and distils off first, leaving ethyl butanoate (b.p. 121 °C) [1]	2
40(a)	Reagents/conditions: Dilute hydrochloric acid (or dilute H ₂ SO ₄), heat under reflux [1] CH ₃ CH ₂ COOCH ₃ (l) + H ₂ O(l) ⇌ CH ₃ CH ₂ COOH(l) + CH ₃ OH(l) [1]	2
40(b)	Reagents/conditions: Aqueous sodium hydroxide (NaOH(aq)), heat under reflux [1] CH ₃ CH ₂ COOCH ₃ + NaOH → CH ₃ CH ₂ COONa + CH ₃ OH [1]	2
40(c)	In alkaline hydrolysis, the hydroxide ion acts as both a nucleophile and a base [1] It immediately deprotonates the carboxylic acid formed to produce a carboxylate anion (CH ₃ CH ₂ COO ⁻) [1] The negatively charged carboxylate ion is resonance-stabilised and repels nucleophilic attack by the alcohol, preventing the reverse reaction [1]	3
40(d)	Add dilute strong mineral acid (e.g. dilute HCl or dilute H ₂ SO ₄) to protonate the carboxylate anion: CH ₃ CH ₂ COO ⁻ + H ⁺ → CH ₃ CH ₂ COOH [1]	1
41(a)	Alcohol N contains the CH ₃ -CH(OH)- group [1]	1
41(b)	Carboxylic acid Q must have at least 4 carbons to possess a chiral centre: 2-methylbutanoic acid: CH ₃ CH ₂ -CH*(CH ₃)-COOH (5 carbons) or chiral hydroxy acid [1] Total carbons in ester = 6; if alcohol N has at least 3 carbons (e.g. propan-2-ol, C ₃), acid has 3 carbons (achiral). If alcohol is ethanol (C ₂), acid Q has 4 carbons (achiral). If acid Q is 2-hydroxypropanoic, oxygen count would be 3. Therefore, reconsider chiral acid with 4 carbons: 2-methylpropanoic is achiral. Thus, acid Q must be 2-chloropropanoic? No, formula C ₆ H ₁₂ O ₂ . For C ₆ H ₁₂ O ₂ : ester M forms alcohol N and salt P. If alcohol N is ethanol (C ₂ H ₅ OH, positive iodoform!), acid has 4 carbons: CH ₃ CH ₂ CH ₂ COOH or (CH ₃) ₂ CHCOOH (both achiral). If alcohol N is propan-2-ol (C ₃ , positive iodoform!), acid Q is propanoic (achiral). If ester is chiral: 1-methylpropyl ethanoate: alcohol is butan-2-ol (chiral!) and acid is ethanoic [2]	2
41(c)	Structure: CH ₃ COOCH(CH ₃)CH ₂ CH ₃ (sec-butyl ethanoate / 1-methylpropyl ethanoate) [1] Systematic name: 1-methylpropyl ethanoate [1]	2
42(a)	O-H stretching vibration of carboxylic acid [1] 2500 - 3000 cm ⁻¹ (overlapping with C-H stretching at ~2950 cm ⁻¹) [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
42(b)	Carboxylic acids form exceptionally strong hydrogen-bonded cyclic dimers [1] The hydrogen bonds weaken and perturb the O-H bond to a much greater and more varied extent across the crystal/liquid lattice than in alcohols [1]	2
42(c)	Ethyl ethanoate lacks an O-H group and shows NO absorption between 2500 and 3600 cm ⁻¹ (shows only sharp C-H peaks and strong C=O at ~1735 cm ⁻¹) [1]	1
43(a)	[CH ₃ -CO] (ethanoyl / acylium cation, m/z = 15 + 28 = 43) [1]	1
43(b)	[CH ₃ CH ₂ O] or [OCH ₂ CH ₃] (ethoxide cation, m/z = 29 + 16 = 45) [1]	1
43(c)	[CH ₃ COOCH ₂] (loss of methyl radical •CH ₃ from ester, m/z = 88 - 15 = 73) [1]	1
44(a)	Reagent: Bromine (Br ₂) [1] Condition: Ultraviolet (UV) light / sunlight [1] Mechanism: Free-radical substitution [1]	3
44(b)	Reagent: Aqueous sodium hydroxide (NaOH(aq)) [1] Condition: Heat under reflux (nucleophilic substitution) [1]	2
44(c)	Acidified potassium dichromate(VI) (K ₂ Cr ₂ O ₇ + dilute H ₂ SO ₄) [1] Heat under reflux [1]	2
44(d)	Catalyst: Concentrated sulfuric acid (conc. H ₂ SO ₄), heat under reflux [1] CH ₃ COOH + CH ₃ CH ₂ OH <=> CH ₃ COOCH ₂ CH ₃ + H ₂ O [1]	2
45(a)	B (Ethyl ethanoate) [1]	1
45(b)	Products: Ethanoic acid and ethanol [1] Ethanoic acid is a carboxylic acid, so it reacts with Na ₂ CO ₃ to liberate CO ₂ ; ethanol is the only primary alcohol that gives a positive tri-iodomethane test [1]	2
46(a)	Moles of NaOH = 0.02000 x 0.100 = 2.00 x 10 ⁻³ mol [1]	1
46(b)	Dicarboxylic acid reacts in 1:2 ratio with NaOH -> moles of acid = (2.00 x 10 ⁻³) / 2 = 1.00 x 10 ⁻³ mol [1]	1
46(c)	Total moles in 250 cm ³ = 1.00 x 10 ⁻³ x 10 = 1.00 x 10 ⁻² mol [1] Mr = mass / moles = 1.46 / 0.0100 = 146.0 [1]	2
46(d)	Formula mass: 2 x COOH (90) + n x CH ₂ (14n) = 146 -> 14n = 56 -> n = 4 [1] Systematic name: Hexanedioic acid (adipic acid) [1]	2
47(a)	Reagents/conditions: Steam (H ₂ O(g)) and concentrated H ₃ PO ₄ catalyst at 300 °C and 60 atm [2]	2
47(b)	CH ₃ COCl + CH ₃ CH(OH)CH ₃ -> CH ₃ COOCH(CH ₃) ₂ + HCl [1] Steamy / misty fumes of HCl evolved [1]	2
47(c)	Correct skeletal formula of 1-methylethyl ethanoate [1]	1
48(a)	B (Sodium carbonate) [1]	1
48(b)	2H ⁺ (aq) + CO ₃ ²⁻ (aq) -> H ₂ O(l) + CO ₂ (g) (or RCOOH + HCO ₃ ⁻ -> RCOO ⁻ + H ₂ O + CO ₂) [1]	1
49(a)	CH ₃ CONH ₂ + H ₂ O + HCl -> CH ₃ COOH + NH ₄ Cl [1]	1
49(b)	CH ₃ CONH ₂ + NaOH -> CH ₃ COONa + NH ₃ [1] Ammonia gas turns damp red litmus paper blue [1]	2
49(c)	The lone pair of electrons on the nitrogen atom is strongly delocalised into the carbonyl pi system [1] This significantly reduces the partial positive charge on the carbonyl carbon (less electrophilic) and gives the C-N bond partial double bond character, increasing activation energy for hydrolysis [1]	2
50(a)	Carboxylic acid group (-COOH) [1]	1
50(b)	Contains the CH ₃ -CH(OH)- group [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
50(c)	Formula C ₄ H ₈ O ₃ with -COOH and CH ₃ -CH(OH)-: CH ₃ -CH(OH)-CH ₂ -COOH [2] Systematic name: 3-hydroxybutanoic acid [1]	3
50(d)	Displayed formula of 3-oxobutanoic acid: CH ₃ -C(=O)-CH ₂ -COOH (showing C=O ketone, CH ₂ , and COOH) [1]	1